

11/6, 17, 18, 19, 20

Ajanta

Page No.

Date

Random Forest

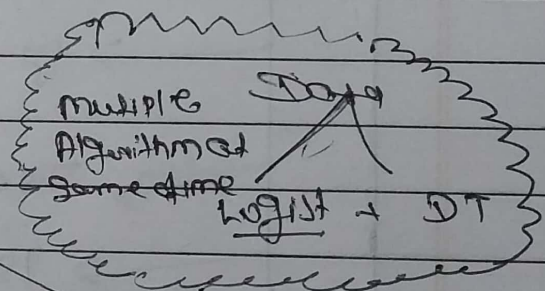
Ensemble Learning Method.

- ① Bagging method: Solves overfitting issue.
- ② Boosting method: Solve underfitting issue.

Bagging: Random Forest.

Boosting: AdaBoost, Gradient Boost.

X GBoost



DT → Log → DT

Bagging: Random Forest.

Boosting: AdaBoost, Gradient Boost, XGBoost.

So much powerful to use real world scenario.

Current ML process.

Training Data → ML Algo DT

overfits

ML Model

pred

yes/no.

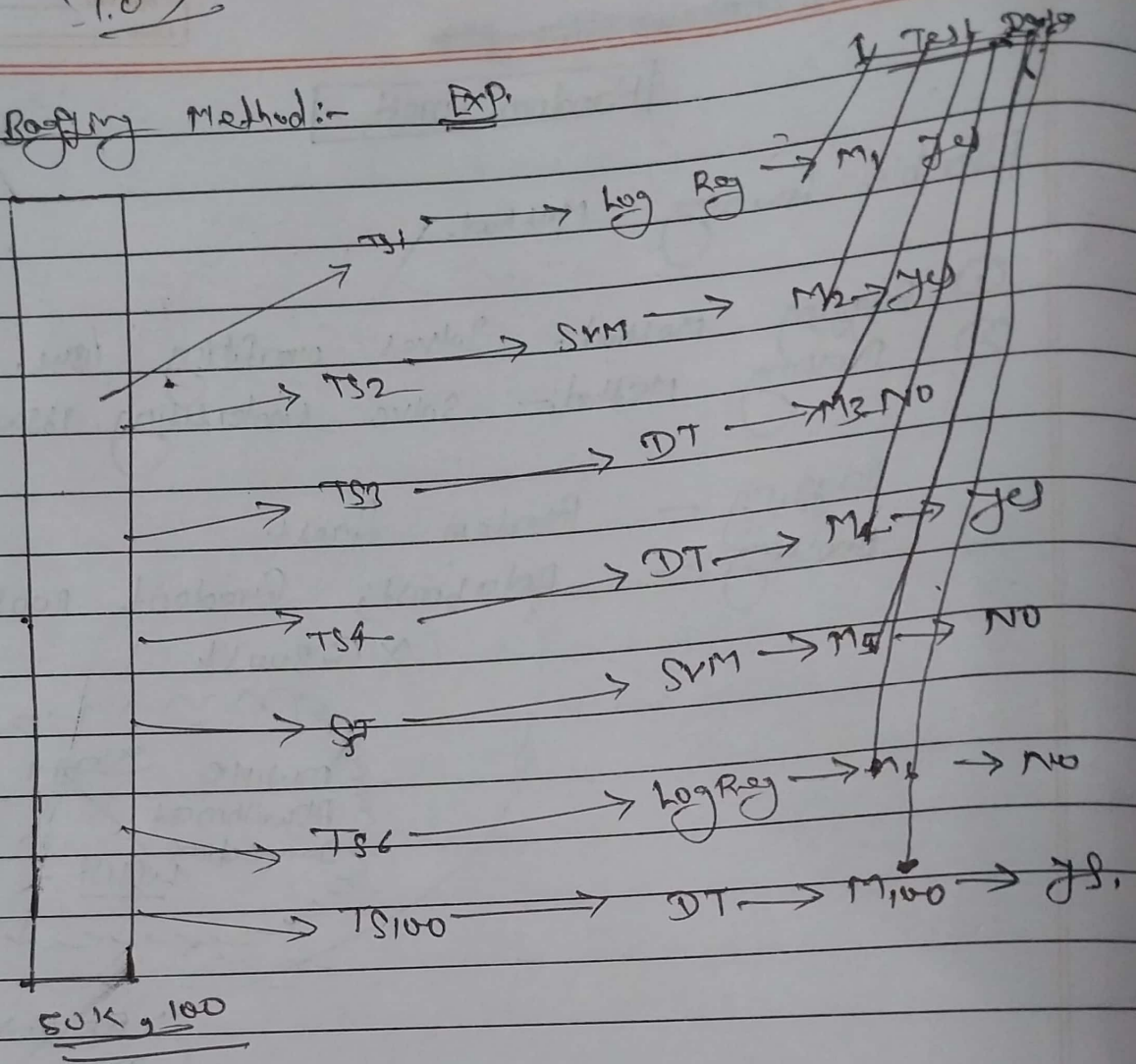
Tree

FI

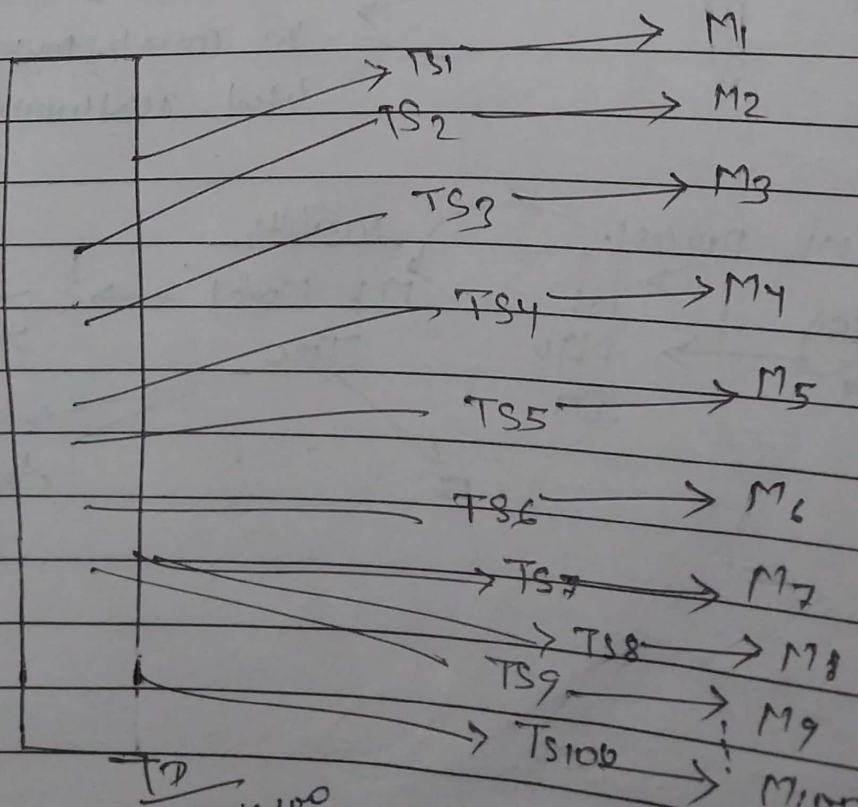
single point dependency.

11.07

Bagging Method:- Exp.



TS₁ = Training sample.



This Particular version of Bagging method was found to perform the best compared to any other combination.

So this version was given a name, which is "Random Forest Algorithm."

In Random Forest only decision tree works.

because it is nothing but the combination of 100 decision Tree.

* Handling for imbalanced data.

1. > Oversampling (SMOTE)
2. > Undersampling (Random Under Sampler).
3. > Use Tree Based Algorithms.

Most Preferred

Notes:—

Never use accuracy score in the case of imbalanced data. Use roc auc score instead.

Notes:—

Precision:—

Recall

F1

Precision:—

Networking

Notes:- In classification data null values filled with ~~*mode*~~
~~*mode*~~

monthly charges and total charges of target column.

Important Notes:-

If any column in dataset is numeric but it is throwing error while change of dtype in, means there is problem with the column if it has String ()

So,

change it by pd.to_numeric.